



\huge National Autonomous University of Mexico

\huge Faculty of Engineering

Hola!

This notebook is part of a series of geoscience applications designed to demonstrate practical approaches for solving problems using computational methods. It serves as a hands-on guide for understanding key concepts in energy-related subsurface systems through applied problem-solving rather than purely theoretical development. The focus is on building intuition and technical skills for modeling and analysis in real-world contexts, particularly within subsurface energy systems. These include petroleum reservoirs, geothermal resources, carbon storage and sequestration, and mineral exploration systems. By integrating fundamental principles with computational tools, the series aims to support learners in bridging the gap between geoscience theory and engineering practice, enabling more effective analysis and decision-making in complex subsurface environments.

Authors:

Dr. Moisés Velasco-Lozano

\huge Darcy's Law

1. Introduction

The flow of water or any other liquid through a porous medium is governed by an empirical relationship derived from experimental observation. This relationship, known as Darcy's Law, represents one of the fundamental principles used to describe fluid flow through porous materials and forms the basis of many applications in reservoir engineering, hydrogeology, soil mechanics, and subsurface flow modeling.

In 1856, the French engineer Henry Darcy conducted a series of experiments to investigate the movement of water through columns packed with unconsolidated sand. His experimental apparatus consisted of a vertical column filled with porous material through which water was allowed to flow under a controlled hydraulic-head difference. By measuring the volumetric flow rate and relating it to the geometry of the porous medium and the hydraulic conditions imposed across the column, Darcy identified a systematic relationship between these variables.

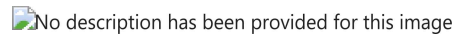
Darcy observed that the volumetric flow rate, q , was directly proportional to the cross-sectional area, A , of the porous medium and to the difference in hydraulic head between the inlet and outlet, while being inversely proportional to the length, L , over which the pressure or hydraulic-head change occurred.

These observations led to the mathematical expression now known as Darcy's Law:

$$q = -KA \frac{\Delta h}{L}$$

where q is the volumetric flow rate, K is the hydraulic conductivity of the porous medium, A is the cross-sectional area available for flow, Δh is the hydraulic-head difference, and L is the length of the porous medium.

The following figure shows the typical configuration of the experiment:



The negative sign indicates that fluid naturally flows in the direction of decreasing hydraulic potential or hydraulic head.

Darcy's experiments demonstrated that the ability of a fluid to move through a porous material depends not only on the applied driving force but also on the properties of the porous medium and the fluid itself. This concept was later generalized for petroleum-reservoir applications by introducing permeability, k , and fluid viscosity, μ , allowing Darcy's Law to be expressed in terms of the pressure gradient.

Today, Darcy's Law provides the foundation for describing single-phase and multiphase flow through reservoir rocks and is extensively used to model the movement of oil, water, and gas in subsurface formations.

2. Numerical solution

Incompressible fluid

$$c_f = \text{constant}$$

Slightly compressible fluid

$$V = V_{ref} [1 + c_f(p_{ref} - p)]$$

Compressible fluid

$$c_g = \frac{1}{p} - \frac{1}{z} \left(\frac{\partial z}{\partial p} \right)_T$$

2.1 Import libraries

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
```

2.2 Inputs

```
In [2]: # Conversion factors

md_to_m2 = 9.869233e-16      # millidarcy -> m²
cp_to_pas = 1.0e-3          # cP -> Pa·s
psi_to_pa = 6894.757         # psi -> Pa
cm2_to_m2 = 1.0e-4          # cm² -> m²
m3s_to_m3day = 86400        # m³/s -> m³/day

# Reference conditions

k_ref = 100                  # mD
A_ref = 100                  # cm²
mu_ref = 1.0                 # cP
L_ref = 1.0                  # m
```

2.3 Calculation and plotting

```
In [3]: # Darcy equation
# q = k A ΔP / (μ L)
def darcy_flow(k_md, A_cm2, mu_cp, dp_psi, L_m):

    k = k_md * md_to_m2
    A = A_cm2 * cm2_to_m2
```

```

mu = mu_cp * cp_to_pas
dp = dp_psi * psi_to_pa

q = (k * A * dp) / (mu * L_m)

return q * m3s_to_m3day    # m³/day

# Figure
fig, axes = plt.subplots(2, 2, figsize=(13, 9))

# =====
# 1. Effect of permeability
# =====

dp = np.linspace(0, 2000, 100)

k_values = [50, 100, 200] # mD

for k in k_values:

    q = darcy_flow(
        k_md=k,
        A_cm2=A_ref,
        mu_cp=mu_ref,
        dp_psi=dp,
        L_m=L_ref
    )

    axes[0, 0].plot(
        dp,
        q,
        linewidth=2,
        label=f'k = {k} mD'
    )

axes[0, 0].set_title('Effect of Permeability')
axes[0, 0].set_xlabel('Pressure Drop, ΔP (psi)')
axes[0, 0].set_ylabel('Flow Rate, q (m³/day)')
axes[0, 0].legend()
axes[0, 0].grid(alpha=0.3)

# =====
# 2. Effect of cross-sectional area
# =====

A_values = [50, 100, 200] # cm²

for A in A_values:

    q = darcy_flow(
        k_md=k_ref,
        A_cm2=A,
        mu_cp=mu_ref,
        dp_psi=dp,
        L_m=L_ref
    )

    axes[0, 1].plot(
        dp,
        q,
        linewidth=2,
        label=f'A = {A} cm²'
    )

```

```

axes[0, 1].set_title('Effect of Cross-Sectional Area')
axes[0, 1].set_xlabel('Pressure Drop,  $\Delta P$  (psi)')
axes[0, 1].set_ylabel('Flow Rate, q (m3/day)')
axes[0, 1].legend()
axes[0, 1].grid(alpha=0.3)

# =====
# 3. Effect of viscosity
# =====

mu_values = [0.5, 1.0, 2.0] # cP

for mu in mu_values:

    q = darcy_flow(
        k_md=k_ref,
        A_cm2=A_ref,
        mu_cp=mu,
        dp_psi=dp,
        L_m=L_ref
    )

    axes[1, 0].plot(
        dp,
        q,
        linewidth=2,
        label=f' $\mu = \{mu\}$  cP'
    )

axes[1, 0].set_title('Effect of Fluid Viscosity')
axes[1, 0].set_xlabel('Pressure Drop,  $\Delta P$  (psi)')
axes[1, 0].set_ylabel('Flow Rate, q (m3/day)')
axes[1, 0].legend()
axes[1, 0].grid(alpha=0.3)

# =====
# 4. Effect of pressure drop
#
# Here length varies so that we can clearly see the influence
# of different pressure drops.
# =====

L = np.linspace(0.2, 5.0, 100)

dp_values = [500, 1000, 2000] # psi

for dp_value in dp_values:

    q = darcy_flow(
        k_md=k_ref,
        A_cm2=A_ref,
        mu_cp=mu_ref,
        dp_psi=dp_value,
        L_m=L
    )

    axes[1, 1].plot(
        L,
        q,
        linewidth=2,
        label=f' $\Delta P = \{dp\_value\}$  psi'
    )

axes[1, 1].set_title('Effect of Pressure Drop')

```

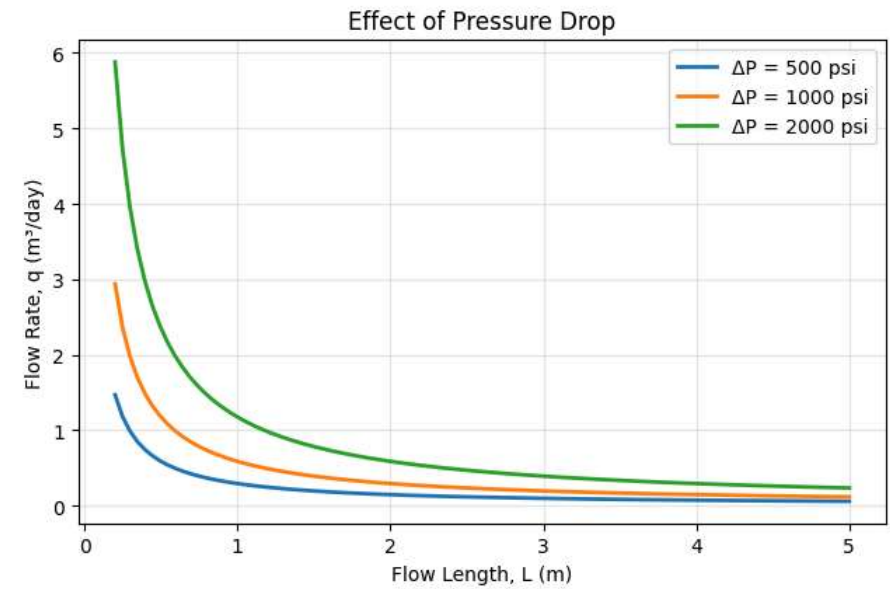
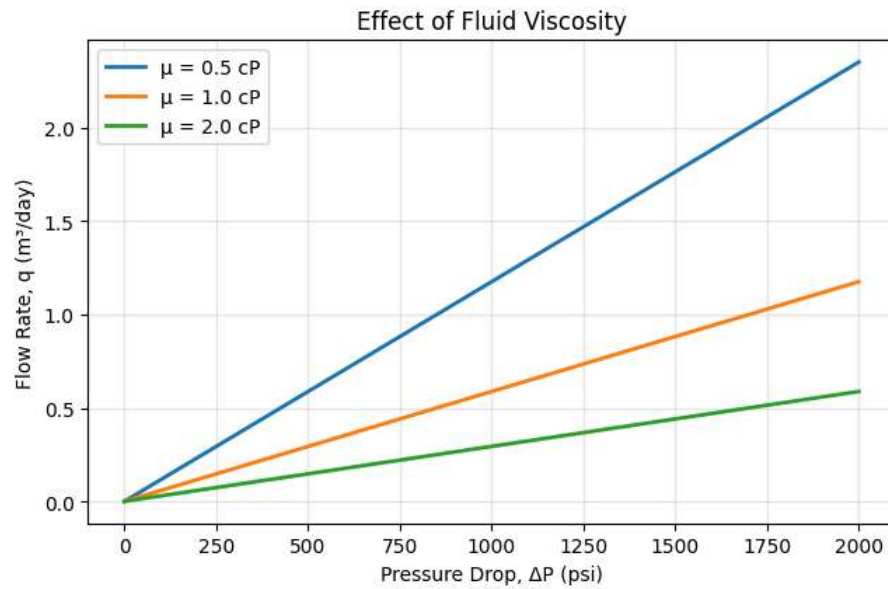
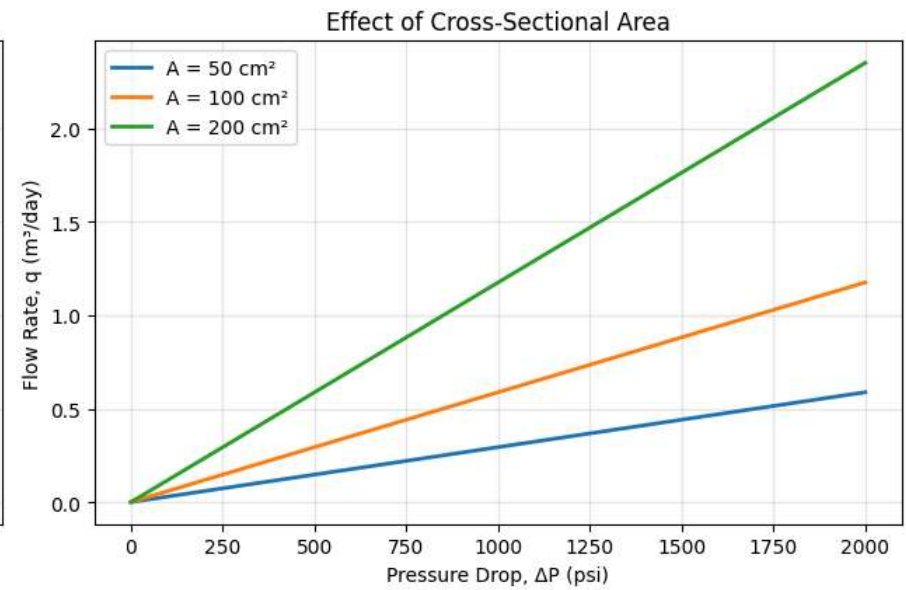
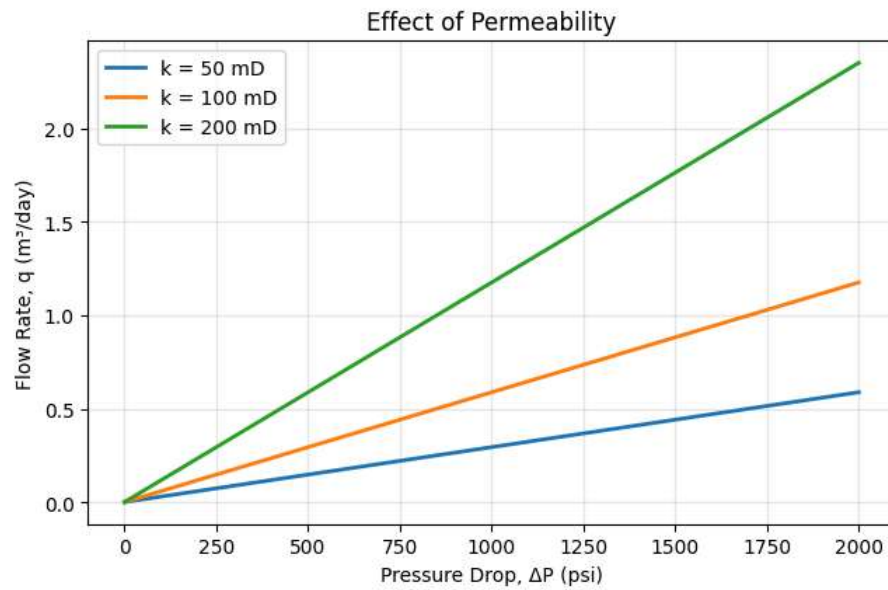
```
axes[1, 1].set_xlabel('Flow Length, L (m)')
axes[1, 1].set_ylabel('Flow Rate, q (m³/day)')
axes[1, 1].legend()
axes[1, 1].grid(alpha=0.3)

# =====
# General formatting
# =====

fig.suptitle(
    "Sensitivity of Darcy's Law Parameters",
    fontsize=16,
    fontweight='bold'
)

plt.tight_layout()
plt.show()
```

Sensitivity of Darcy's Law Parameters



3. Conclusions and highlights

In []:

4. Comments

I hope this notebook is useful in your journey of coding applied to petroleum engineering and geosciences. I am a researcher interested in Machine learning for subsurface energy systems including oil & gas, carbon capture and storage, geothermal, and minerals. Please reach out!

Dr. Moisés Velasco-Lozano
Associate Professor
Petroleum Engineering Department
National Autonomous University of Mexico (UNAM)

mvelasco@ingenieria.unam.edu
mvelasco@unam.mx

In []: